

Distributed Control and Communication Framework for Connectivity Preservation in Flow-Driven Robot Fleets

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Abstract—We study the deployment of underwater robot fleets in large, flow dominated environments where localization is denied, communication is limited, and energy is scarce. The robots need to travel long distances, detect emerging targets, and stay connected using only local information. To meet these challenges we design a distributed framework that exploits natural drift and activates control only when required. The method begins with the robots deployed as a fully connected network. Using neighbor-to-neighbor message exchange, robots prune redundant edges by constructing a spanning-tree coordination structure that spans the entire team. Upon this structure we construct a distributed control Lyapunov function, control barrier function (CLF-CBF) framework. Each robot enforces barrier constraints to maintain the necessary connections, while redundant links are discarded when alternative paths emerge. When a target is discovered, this framework ensures the detecting robot attends the target, while the team reconfigures to provide a connected formation. This approach guarantees long term connectivity with very low control effort, and enables collective response to dynamic tasks. Simulations validate the framework and demonstrate substantial edge reduction and low control effort while preserving end-to-end connectivity under bounded disturbances.

I. INTRODUCTION

We are interested in deploying a fleet of underwater robots into large-scale, challenging environments such as the deep ocean and under-ice cavities, where localization is denied, communication bandwidth is limited, and the environment is dominated by dynamic flows and eddies [1]. These robots must traverse long distances on limited energy budgets, discover emerging regions of interest, sample the environment, and relay data, all while relying on intermittent and bandwidth-limited communication [2], [3].

The central challenge is to design strategies that enable the fleet to (i) efficiently cover a large spatial domain to discover emerging targets while maintaining network connectivity, and (ii) realize a globally connected configuration using only localized information and limited communication.

Our proposed approach is to deploy the robots together and then allow them to drift with the prevailing flow. Control effort is only applied when it is necessary. This includes situations when the robots risk drifting outside a prescribed region or when they risk losing connectivity. In this way the formation disperses while staying connected, and the group

can also reconfigure when a new target location is discovered or generated [4]. This strategy is bold, but it is supported by recent progress. Previous work has shown that surface vehicles can drift with the flow while using minimal control to remain in gyre like structures [5], or track the periodicity of revisits via intermittent control triggered by prescribed conditions [6]. In addition, broadly adopted models describe the uncontrolled motion of objects in water as drift combined with Brownian type diffusion [7]–[9], which align with observation on underwater robots [10], [11]. This indicates that our fleet will approximately follow the main direction of the flow while also dispersing stochastically.

Although dispersion promotes coverage of large task spaces, two challenges remain open. The first is how to maintain connectivity. The second is how to enable re-configuration around targets while working with unreliable communication that cannot support centralized control. Prior work on connectivity maintenance and target surrounding has emphasized the importance of graph theoretic methods [12], [13]. Control barrier functions (CBFs) have received attention for their energy efficiency in this setting. For example, CBFs have been applied to the Fiedler value of the network graph in order to maintain connectivity [14]. This approach is effective but it requires centralized and computationally expensive eigenvalue calculations. Other work has applied distributed CBF formulations that keep robots pairwise within communication range [15], [16]. More recently, a hybrid framework was introduced that combines distributed CBFs with a graph theoretic tool, where each robot interacts only with a limited set of neighbors defined by a minimum spanning tree (MST) [17]. This solution is promising since each robot only applies CBF to itself and to selected neighbors, but it still requires generating a minimum spanning tree that depends on global information.

In our case a feasible framework must combine two ingredients. The first is a distributed neighbor selection strategy so that each robot chooses a limited set of neighbors to remain within communication range. The second is a local control law applied only to these neighbors. Together these two features must guarantee global connectivity. Work on spanning trees has shown that distributed verification methods exist, but the actual construction of a minimum spanning tree is classically centralized [18]–[20]. Progress in fully distributed protocol for exact MST construction [21], [22] remain communication-intensive. As a result, several approximate or near-optimal distributed methods have been proposed, but they do not achieve exact solutions [23].

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We take advantage of two physical features of our problem. First, the robots are deployed together, and the initial configuration forms a fully connected graph. Second, the purpose of a spanning tree is to reduce long range connections. A robot should drop links to distant neighbors whenever a closer neighbor can maintain connectivity at lower control cost. Recent progress [24] has also shown that robots can verify whether an edge is critical, meaning that breaking it would break the network connectivity, using distributed procedures that rely only on very limited data sharing. Based on these observations we propose a distributed method that relies only on one-hop neighbor messages. Robots exchange neighbor-to-neighbor messages during a topology-discovery phase (as in [24]) and then construct a sparse spanning-tree coordination structure via a deterministic parent-selection rule; all redundant edges are pruned locally. Unlike [24], where critical edges are detected as a diagnostic for link removal, our method uses hop-distance discovery only as a graph-information primitive and builds a provably connected pruned structure that directly defines which links are maintained by the CLF CBF controller. The result is a sparse and tree-like structure that spans the entire network. Once this structure is established, we construct a distributed control Lyapunov function–control barrier function (CLF–CBF) framework upon it. Each robot maintains only the necessary connections by enforcing CBF constraints, and redundant links are pruned whenever new edges provide alternative paths. When a target is detected, the framework drives the robot that discovers it toward the target, while the rest of the team reconfigures to form a supporting network from the bottom up. As a result, the team preserves connectivity, expands to maintain coverage, and collectively shifts toward the target location.

Simulations demonstrate that the method rapidly prunes redundant links to a spanning-tree coordination structure while preserving end-to-end connectivity and reaching the target under flow-driven dispersion. Over deployment time the spanning structure produced by this distributed pruning strategy remains sparse and tree-like, reducing communication redundancy while preserving connectivity. This shows both its long term efficiency and its robustness in flow dominated and resource constrained conditions.

This paper is organized as follows. Sec. II introduces the problem. Sec. III presents our methodology of achieving the structure and moving towards the desired location. Sec. IV certifies the preservation of the communication links over the graph. Sec. V explains the experimental setup and Sec. VI demonstrates the results of the proposed method via simulated results. Sec. VII concluded this work.

II. BACKGROUND AND PROBLEM FORMULATION

We now formalize the problem. This section introduces the robot dynamics model, states the assumptions, defines the graph-theoretic notation, and presents the formal problem statement.

A. Advection-diffusion model [7]–[9]

We model a fleet of N robots operating in $\mathbb{R}^n$ (with $n = 2$, used for visualization in our simulations) released together in the water, their collective motion can be decomposed into a drift component induced by an ambient flow field and a stochastic dispersion component due to unexpected turbulence and small perturbations. Let $x_i(t) \in \mathbb{R}^n$ denote the position of the i -th robot at time t , where $i \in \{1, 2, \dots, N\}$, and let

$$\bar{x}(t) = \frac{1}{N} \sum_{i=1}^N x_i(t)$$

be the center of mass of the group.

The evolution of the center of mass is governed primarily by the advective flow field, i.e.,

$$\dot{\bar{x}}(t) \approx f_{\text{flow}}(\bar{x}(t), t),$$

where f_{flow} is the background velocity field. Each individual object then deviates from $\bar{x}(t)$ due to unresolved small-scale turbulence and fluctuations, which can be modeled as Brownian motion:

$$\begin{aligned} dx_i(t) &= (f_{\text{flow}}(x_i(t), t) + v_i(t))dt + \sigma_{\text{diff}}dW_i(t) \\ \dot{v}_i(t) &= u_i(t) - \gamma v_i(t) \end{aligned} \quad (1)$$

where $v_i(t) \in \mathbb{R}^n$ is the velocity of each individual robot, $u_i(t) \in \mathbb{R}^n$ is the thrust, $\gamma > 0$ is a velocity damping factor, and $W_i(t)$ is a Wiener process capturing diffusion (with intensity σ_{diff}).

Thus the group drifts together with the flow through the motion of $\bar{x}(t)$, while individuals spread stochastically around the drifting center due to the independent diffusion terms.

As an example, the flow field is a base current plus a spatially varying swirl:

$$f_{\text{flow}}(x, t) = v_{\text{base}}(t) + A_{\text{swirl}} \begin{bmatrix} \sin\left(\frac{x_2+t}{s_{\text{swirl}}}\right) \\ \cos\left(\frac{x_1+0.3t}{s_{\text{swirl}}}\right) \end{bmatrix}$$

For two robots i and j , the flow difference is generally nonzero when $x_i \neq x_j$:

$$f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t) = A_{\text{swirl}} \begin{bmatrix} \sin\left(\frac{x_{i,2}+t}{s_{\text{swirl}}}\right) - \sin\left(\frac{x_{j,2}+t}{s_{\text{swirl}}}\right) \\ \cos\left(\frac{x_{i,1}+0.3t}{s_{\text{swirl}}}\right) - \cos\left(\frac{x_{j,1}+0.3t}{s_{\text{swirl}}}\right) \end{bmatrix} \neq 0$$

We assume that the ambient flow is spatially smooth and its differential drift is bounded:

$$\|f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)\| \leq L_f \|x_i - x_j\|, \quad \forall x_i, x_j, \quad \forall t \geq 0.$$

If the flow were spatially uniform, it would cancel in the relative dynamics. This assumption explicitly captures the non-uniform, spatially varying regime by bounding how much the flow can differ across inter-robot distances. For the swirl flow field, one can take this flow as globally lipschitz in x with a lipschitz constant $L_f \approx A_{\text{swirl}}/s_{\text{swirl}}$.

B. Control-affine model

For controller synthesis and theoretical analysis, we now abstract the above simulation dynamics into a compact control-affine model on x_i :

$$\dot{x}_i(t) = f_{\text{flow}}(x_i(t), t) + u_i(t) + d_i(t), \quad \|d_i(t)\| \leq \bar{d}. \quad (2)$$

With the control input $u_i(t)$ set to zero throughout the deployment time window, the robots will be passively advected by the background flow while gradually dispersing due to stochastic perturbations. Here the unmodeled effects of drag, turbulence and stochastic perturbations are captured by a bounded disturbance term:

$$\|d_i(t)\| \leq \bar{d}, \quad \forall i \in \mathcal{V}, \quad \forall t \geq 0.$$

This term corresponds to the combined effect of Brownian motion, transient velocity dynamics, and model uncertainties that are abstracted away in the reduced model. We present two different dynamics descriptions because they serve different purposes. The advection–diffusion model supports physical interpretation and simulation realism, while a control-affine representation is used for controller design and theoretical analysis in Section IV. The flow field is evaluated at each robot’s position and therefore is not a common input in relative coordinates. Defining the relative state $z_{ij} \triangleq x_i - x_j$, we have

$$\begin{aligned} \dot{z}_{ij} &= \underbrace{(f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t))}_{\triangleq \Delta f_{\text{flow}}(x_i, x_j, t)} + (u_i - u_j) + (d_i - d_j). \end{aligned} \quad (3)$$

Under intermittent communication, robot i does not have access to the instantaneous neighbor control $u_j(t)$. Following Assumption 3, we write

$$u_j(t) = \hat{u}_j(t) + e_{ij}(t), \quad \|e_{ij}(t)\| \leq \bar{e}, \quad (4)$$

where $\hat{u}_j(t)$ is the most recently received neighbor-control estimate and $e_{ij}(t)$ captures staleness. Substituting (4) into (3) yields the implementable form used in the robust CBF constraints:

$$\dot{z}_{ij} = \Delta f_{\text{flow}}(x_i, x_j, t) + (u_i - \hat{u}_j) + (d_i - d_j) - e_{ij}. \quad (5)$$

So, the flow contribution cancels only in the special case of spatially uniform flow. In our setting, spatial variation of f_{flow} is explicitly bounded, which is a quantity that enters the connectivity and invariance analysis.

The time derivative of the squared inter-robot distance can be written as

$$\dot{z}_{ij}^2 = 2z_{ij}^\top \left((f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)) + (v_i - v_j) \right)$$

This expression makes explicit that the flow contributes through the *difference term* $f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)$.

Notice that the choice of a two-dimensional environment is made solely for ease of visualization. The developed method is dimension-agnostic.

C. Assumptions on communication, control and connectivity

Assumption 1 (Bounded actuation and communication). *At time t , the communication network is an undirected disk graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ with range $R_{\text{max}} > 0$:*

$$(i, j) \in \mathcal{E}_t \iff \|x_i(t) - x_j(t)\| \leq R_{\text{max}}. \quad (6)$$

Each robot i communicates only with its neighbors in its neighboring set

$$\mathcal{N}_i(t) \triangleq \{j \in \mathcal{V} : \|x_i(t) - x_j(t)\| \leq R_{\text{max}}\}, \quad (7)$$

using only neighbor communication without any central coordination. Each robot has a unique identifier to index neighbor messages consistently. The initial graph $\mathcal{G}_0$ is connected. We assume robot IDs are drawn from $\{1, \dots, |\mathcal{V}|\}$ and that robots know $n = |\mathcal{V}|$ to index the hop-distance vectors ω_i .

For each robot i and all $t \geq 0$, the control is bounded as $\|u_i(t)\| \leq u_{\text{max}}$. The unmodeled disturbance is bounded by $\|d_i(t)\| \leq \bar{d}$. The ambient flow field $f_{\text{flow}}(x, t)$ is Lipschitz in space with constant L_f over the operating domain, i.e., $\|f_{\text{flow}}(x, t) - f_{\text{flow}}(y, t)\| \leq L_f \|x - y\|$ for all relevant x, y, t .

Assumption 2 (Feasible safety & connectivity margins). *There exist margins $\varepsilon_s > 0$ and $\varepsilon_c > 0$ such that the required inter-robot separation and communication constraints are simultaneously feasible:*

$$0 < d_{\text{min}} + \varepsilon_s < R_{\text{max}} - \varepsilon_c.$$

Moreover, the available control authority is sufficient to counter worst-case relative disturbance over the communication range, i.e.,

$$u_{\text{max}} > \bar{d} + L_f R_{\text{max}}.$$

Assumption 3 (Bounded neighbor-control staleness on maintained links). *For each maintained coordination link $(i, j) \in \mathcal{E}_{\text{topo}}(k)$, robot i receives from robot j a timestamped packet containing its most recently applied control. Let*

$$\hat{u}_j(t) \triangleq u_j(t - \tau_{ij}(t))$$

denote the locally available estimate at robot i , where the staleness satisfies $\tau_{ij}(t) \leq \bar{\tau}$ for all t while (i, j) is maintained. Further assume a bounded control slew rate $\|\dot{u}_j(t)\| \leq \bar{v}$ for all j, t . Then the neighbor-control mismatch $e_{ij}(t) \triangleq u_j(t) - \hat{u}_j(t)$ is bounded as

$$\|e_{ij}(t)\| \leq \bar{e} \triangleq \bar{v} \bar{\tau}.$$

Assumption 4 (Time-scale separation for distributed updates). *The distributed pruning updates run on a faster time scale than the robot motion, so that during one algorithmic iteration the neighbor set can be treated as constant and required neighbor-to-neighbor messages are available for that iteration. Equivalently, topology variations are slow relative to the update period.*

D. Problem Statement

Consider the multi-robot system introduced above with dynamics (2) and communication model (6)–(7). The robots are advected by a spatially varying ambient flow field $f_{\text{flow}}(x, t)$ and are subject to bounded disturbances $d_i(t)$, which together induce natural dispersion. While this dispersion promotes exploration, it can also stretch inter-robot distances and break communication links.

Let $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ denote the physical disk graph induced by the range R_{max} (Assumption 1). During the mission, robots may intentionally sparsify the network by maintaining a time-varying coordination subgraph $\mathcal{G}_{\text{topo}}(t) = (\mathcal{V}, \mathcal{E}_{\text{topo}}(t))$ with $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$.

We formalize the hard safety/connectivity requirements using the constraint sets

$$\mathcal{C}_{\text{safe}} \triangleq \left\{ x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \geq d_{\text{min}}, \forall i \neq j \right\},$$

$$\mathcal{C}_{\text{conn}}(\mathcal{E}) \triangleq \left\{ x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \leq R_{\text{max}}, \forall (i, j) \in \mathcal{E} \right\},$$

and the admissible input set $\mathcal{U} \triangleq \{(u_1, \dots, u_N) : \|u_i\| \leq u_{\text{max}} \forall i\}$ (Assumption 1).

Targets are not known a priori and are detected only when an agent enters its sensing range. During exploration, the team should drift with f_{flow} while remaining connected and reducing communication redundancy. Upon target detection, agents should initiate goal-seeking behavior maintaining end-to-end connectivity across the fleet.

Each agent i has access only to local information: (i) its own state x_i ; (ii) relative states $\{x_j - x_i\}_{j \in \mathcal{N}_i(t)}$ and messages from one-hop neighbors; and (iii) local graph quantities, like two hop connection, are computable through neighbor exchanges. We denote this information set by $\mathcal{I}_i(t)$. No centralized computation, global maps, or spectral quantities are assumed; a spanning-tree coordination structure is constructed distributedly using one-hop neighbor messages. Distributed connectivity-preserving control with topology sparsification. Design distributed policies consisting of (i) a feedback control law $u_i(t) = \kappa_i(\mathcal{I}_i(t))$ for each agent and (ii) a distributed concurrent pruning rule that generates $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$, using only local information, such that for all $t \geq 0$:

- 1) $x(t) \in \mathcal{C}_{\text{safe}}$ and $x(t) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_{\text{topo}}(t))$, and the maintained graph $\mathcal{G}_{\text{topo}}(t)$ remains connected.
- 2) The controller exploits advection and uses minimal actuation.
- 3) The team spreads for coverage during exploration and, while expansion figuring out the current graph structure from local knowledge and periodically constructing a spanning-tree coordination structure and pruning redundant edges using only local information to make a sparse graph network.

The proposed framework in Sec. III provides a fully distributed realization of these objectives via distributed topology sparsification and decentralized control.

III. METHODOLOGY

Our framework couples a *graph layer* with a *control layer*. At discrete update times t_k , the graph layer constructs a sparse connected coordination structure using only one-hop neighbor messages, and prunes all other links. Between update times, the control layer implements a CBF safety filter where a nominal CLF controller drives robots toward task goals, while CBF constraints enforce (i) collision avoidance with nearby robots and (ii) connectivity range maintenance for the maintained coordination links. Although the instantaneous disk graph $\mathcal{G}_t$ varies with advection and diffusion, the connectivity CBF constraints ensure that maintained links stay within range between updates and therefore do not drop from the disk graph.

Notation. Robots evolve under the control-affine model (2). The instantaneous communication disk graph is defined in (6)–(7). Let t_k denote the k -th topology update time. At update k , the graph layer outputs a maintained edge set $\mathcal{E}_{\text{topo}}(k) \subseteq \mathcal{E}_{t_k}$ that forms a spanning tree of $\mathcal{G}_{t_k}$. For robot i , define the *maintained neighbor set* $\mathcal{T}_i(k) \triangleq \{j : (i, j) \in \mathcal{E}_{\text{topo}}(k)\}$. Separately, define a *safety neighbor set* $\mathcal{S}_i(t)$ used only for collision avoidance.

A. Graph layer: distributed spanning-tree pruning

At each topology update time t_k , robots observe their one-hop neighbor sets $\mathcal{N}_i(t_k)$ induced by the disk graph. The objective of the graph layer is to construct a sparse connected coordination structure so that the control layer enforces connectivity constraints on only $|\mathcal{V}| - 1$ links. Our graph-layer design follows the same high-level philosophy as recent distributed link-reduction methods [24]. Robots make pruning decisions using only one-hop neighbor messages and without centralized computation. In particular, the distributed hop-distance propagation in (8) computes shortest-path hop distances using only one-hop message exchange. Our pruning uses only distances-to-root equation (9), which is sufficient to construct a connected spanning tree. We do not require explicit critical-edge identification for correctness, since any spanning tree preserves end-to-end connectivity and retains all bridges of the original connected graph.

1) *Distributed hop-distance discovery:* We adopt the distributed hop-distance discovery mechanism from [24] as a lightweight graph-information primitive. Each robot i maintains a hop-distance vector $\omega_i(\cdot) \in (\mathbb{N} \cup \{\infty\})^{|\mathcal{V}|}$, where $\omega_{i,j}(q)$ is robot i 's estimate at round q of the shortest-path hop distance from i to node j . Initialize

$$\omega_{i,i}(0) = 0, \quad \omega_{i,j}(0) = \infty \text{ for } j \neq i.$$

At each synchronous round q , robot i receives $\omega_\ell(q)$ from neighbors $\ell \in \mathcal{N}_i(t_k)$ and updates, for all $j \in \mathcal{V}$,

$$\omega_{i,j}(q+1) = \min \left\{ \omega_{i,j}(q), 1 + \min_{\ell \in \mathcal{N}_i(t_k)} \omega_{\ell,j}(q) \right\}. \quad (8)$$

Under Assumption 4, the iteration converges to exact hop distances within at most $|\mathcal{V}| - 1$ rounds, i.e., $\omega_{i,j}(q) = d(i, j)$ for all i, j once $q \geq |\mathcal{V}| - 1$.

For our pruning method we only require distances-to-root: fixing a root r (e.g., smallest ID), define

$$\delta_i \triangleq \omega_{i,r}(q^*), \quad (9)$$

where $q^* \geq |\mathcal{V}| - 1$ is any round after convergence.

2) *Parent selection and local keep/prune rule:* After the hop-distance discovery converges, each node uses $\delta_i = \omega_{i,r}$ from (9) to select a parent. Then each non-root node selects a parent from neighbors that are one hop closer to the root:

$$C_i \triangleq \{j \in \mathcal{N}_i(t_k) : \delta_j = \delta_i - 1\}, \quad p(i) \triangleq \min C_i, \quad i \neq r, \quad (10)$$

and the root has no parent. Each node broadcasts its parent choice $p(i)$ to its neighbors (a single scalar). For each undirected edge $\{i, j\} \in \mathcal{E}_{t_k}$ define the keep indicator

$$K_{ij} \triangleq \mathbf{1}\{p(i) = j \text{ or } p(j) = i\}. \quad (11)$$

The pruned edge set is

$$\mathcal{E}_{\text{topo}}(k) \triangleq \{\{i, j\} \in \mathcal{E}_{t_k} : K_{ij} = 1\}. \quad (12)$$

The parent rule (10) makes every non-root node choose a neighbor that is strictly *closer* in hop distance to the root. Therefore each node has a strictly descending hop-distance chain to the root, and the union of parent edges forms a rooted shortest-path BFS tree over $\mathcal{G}_{t_k}$. This pruning decision is distributed as robot i needs only its local neighbor IDs, the received δ_j , and the received parent IDs $p(j)$ from one-hop neighbors.

Theorem 1 (Distributed spanning-tree pruning). *Assume that the disk graph $\mathcal{G}_{t_k} = (\mathcal{V}, \mathcal{E}_{t_k})$ is connected and Assumption 4 holds so that neighbor sets remain constant during the update window. Then the output $(\mathcal{V}, \mathcal{E}_{\text{topo}}(k))$ is a spanning tree of $\mathcal{G}_{t_k}$: it is connected, acyclic, and has exactly $|\mathcal{V}| - 1$ edges.*

Proof: After convergence of (8), we have $\omega_{i,r}(q^*) = d(i, r)$ for all i , hence $\delta_i = d(i, r)$ by (9). For each $i \neq r$, there exists a shortest path from i to r , therefore there exists a neighbor $j \in \mathcal{N}_i(t_k)$ on that path with $\delta_j = \delta_i - 1$, so $C_i \neq \emptyset$ and $p(i)$ is well-defined. Each non-root node contributes exactly one parent edge $\{i, p(i)\}$ and this edge is kept because $K_{i,p(i)} = 1$; therefore, $|\mathcal{E}_{\text{topo}}(k)| = |\mathcal{V}| - 1$. Along each parent edge, δ decreases by one; therefore, no directed cycle exists in the parent-pointer graph, implying that $(\mathcal{V}, \mathcal{E}_{\text{topo}}(k))$ is acyclic. Finally, following parent pointers from any node strictly decreases δ and must terminate at r , yielding a path from every node to r in $(\mathcal{V}, \mathcal{E}_{\text{topo}}(k))$; hence the graph is connected.

3) *Communication, computation, and memory cost:* Let $\deg(i)$ denote the degree of robot i in $\mathcal{G}_{t_k}$ and let $n = |\mathcal{V}|$. In each hop-distance round (8), robot i broadcasts its length- n vector $\omega_i(q)$ to neighbors and performs $O(\deg(i)n)$ min-updates. The hop-distance stage completes in at most $n - 1$ rounds, requires $O(n)$ memory per robot, and communicates $O(\deg(i)n)$ scalar entries per round. After convergence, each robot broadcasts one scalar parent ID $p(i)$ and prunes incident edges locally using (11).

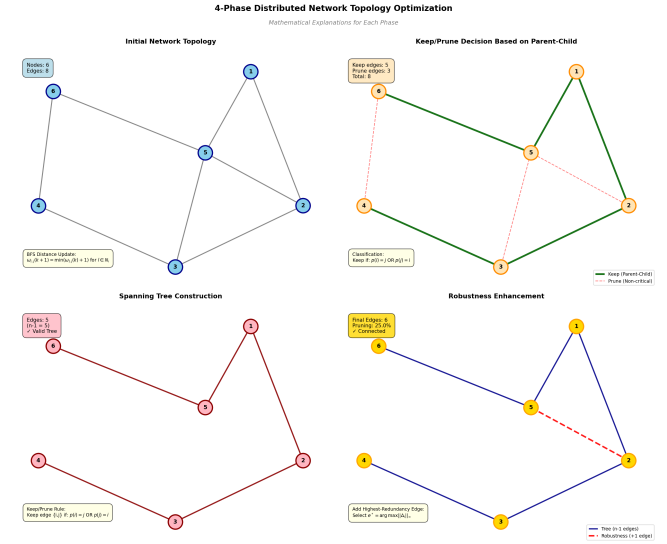


Fig. 1. Four-phase evolution on a 6-node sparse network. Phase 1: initial disk-graph topology. Phase 2: Parent/Child relation. Phase 3: distributed spanning-tree pruning yields $n - 1$ edges. Phase 4 (optional): one additional edge is added to create a single cycle for redundancy.

B. Illustrative example of a sparse graph

In Fig.1 we consider $G = (V, E)$ with $V = \{1, 2, 3, 4, 5, 6\}$, root $r = 1$, and

$$E = \{(1, 2), (2, 3), (3, 4), (1, 5), (5, 6), (4, 6), (2, 5), (3, 5)\}, \\ |V| = 6, \quad |E| = 8.$$

The neighbor sets are

$$N_1 = \{2, 5\}, \quad N_2 = \{1, 3, 5\}, \quad N_3 = \{2, 4, 5\}, \\ N_4 = \{3, 6\}, \quad N_5 = \{1, 2, 3, 6\}, \quad N_6 = \{4, 5\}.$$

Following the distributed hop-distance propagation in [24], robots may compute all-pairs hop distances $\omega_{i,j}$ via local neighbor updates. In our pruning method we only require distances-to-root, i.e., the scalar $\delta_i = \omega_{i,r}$, which is the r -th column of the all-pairs matrix.

Using the converged root distances $\delta_i = \omega_{i,1}$, we obtain

$$\delta_1 = 0, \quad \delta_2 = 1, \quad \delta_5 = 1, \quad \delta_3 = 2, \quad \delta_6 = 2, \quad \delta_4 = 3.$$

Each node selects its parent from neighbors one hop closer to the root:

$$C_i = \{j \in N_i : \delta_j = \delta_i - 1\}, \quad p(i) = \min C_i.$$

This yields

$$p(2) = 1, \quad p(5) = 1, \quad p(3) = 2, \quad p(6) = 5, \quad p(4) = 3, \\ \text{and the maintained tree edges are the parent edges}$$

$$E_T = \{\{1, 2\}, \{1, 5\}, \{2, 3\}, \{3, 4\}, \{5, 6\}\}, \\ |E_T| = 5 = n - 1.$$

Equivalently, for each $\{i, j\} \in E$ we keep the edge iff $p(i) = j$ or $p(j) = i$.

Algorithm 1 Algorithm A: Four-phase distributed pruning with optional redundancy edge

Input: Connected disk graph at update time t_k with neighbor sets $N_i(t_k)$; root r (e.g., smallest ID)

Output: Maintained structure $\mathcal{E}_{\text{topo}}(k)$ (tree), optionally plus one redundancy edge

- 1: **Phase 1 (hop-distance discovery):** run the neighbor propagation (8) for $q = 0, \dots, n-1$ and set $\delta_i \leftarrow \omega_{i,r}(q^*)$ as in (9)
- 2: **Phase 2 (monitor):** if $\delta_i = +\infty$ locally, node i declares itself disconnected and triggers a re-run of the update
- 3: **Phase 3 (core spanning tree):**
- 4: **for all** nodes $i \neq r$ **do**
- 5: $C_i \leftarrow \{j \in N_i(t_k) : \delta_j = \delta_i - 1\}$
- 6: $p(i) \leftarrow \min C_i$
- 7: **end for**
- 8: **Keep edge** $\{i, j\}$ **iff** $p(i) = j$ or $p(j) = i$; let $\mathcal{E}_{\text{topo}}(k) \leftarrow \{\{i, j\} \in \mathcal{E}_{t_k} : K_{ij} = 1\}$
- 9: **Phase 4 (optional redundancy):**
- 10: Select one edge $e^* \in \mathcal{E}_{t_k} \setminus \mathcal{E}_{\text{topo}}(k)$ by a chosen heuristic (e.g., one hop neighbors)
- 11: $\mathcal{E}_{\text{topo}}(k) \leftarrow \mathcal{E}_{\text{topo}}(k) \cup \{e^*\}$
- 12: **return** $\mathcal{E}_{\text{topo}}(k)$

To introduce a single alternate path, we add one edge based on proximity. Mostly one hop neighbors are considered closest but any application dependent heuristic (e.g., $\{2, 5\}$ in Fig. 1) can be used to create one redundant cycle. This step is optional and is not required for the spanning-tree correctness guarantee.

C. Control layer: CLF goal-seeking with CBF safety & connectivity maintenance

Between pruning iterations, each robot runs a CLF-CBF controller under (2). The CLF term drives robots toward task goals, while CBF constraints enforce: (i) *collision avoidance* for all $j \in \mathcal{S}_i(t)$, and (ii) *range maintenance* for all $j \in \mathcal{T}_i(k)$.

1) *Barrier functions (pairwise safety and connectivity):* For any ordered pair (i, j) define $z_{ij} \triangleq x_i - x_j$ and $r_{ij} \triangleq \|z_{ij}\|$. We use the standard squared-distance barriers

$$h_{ij}^{\text{safe}}(x) \triangleq r_{ij}^2 - d_{\min}^2, \quad (13)$$

$$h_{ij}^{\text{conn}}(x) \triangleq R_{\max}^2 - r_{ij}^2, \quad (14)$$

so that $h_{ij}^{\text{safe}} \geq 0$ implies $\|x_i - x_j\| \geq d_{\min}$ and $h_{ij}^{\text{conn}} \geq 0$ implies $\|x_i - x_j\| \leq R_{\max}$.

To handle spatially varying flow and bounded disturbances (Assumptions in Sec. II), we employ linear CBF inequalities. With extended class- $\mathcal{K}$ functions α_s, α_c , we enforce robust CBF inequalities using the locally available neighbor-control estimate $\hat{u}_j$ from Assumption 3. For each $j \in \mathcal{S}_i(t)$:

$$2z_{ij}^\top(u_i - \hat{u}_j) \geq 2L_f r_{ij}^2 + 4\bar{d}r_{ij} - \alpha_s(h_{ij}^{\text{safe}}(x)) + 2r_{ij}\bar{e}, \quad (15)$$

and for each $j \in \mathcal{T}_i(k)$:

$$2z_{ij}^\top(u_i - \hat{u}_j) \leq \alpha_c(h_{ij}^{\text{conn}}(x)) - 2L_f r_{ij}^2 - 4\bar{d}r_{ij} - 2r_{ij}\bar{e}. \quad (16)$$

The additional $\pm 2r_{ij}\bar{e}$ margins ensure correctness under bounded staleness: writing $u_j = \hat{u}_j + e_{ij}$ with $\|e_{ij}\| \leq \bar{e}$ (Assumption 3), the true relative-control term $2z_{ij}^\top(u_i - u_j)$ differs from $2z_{ij}^\top(u_i - \hat{u}_j)$ by at most $2r_{ij}\bar{e}$.

2) *CLF objective and QP-based safety filter:* Let $x_i^{\text{goal}}(t)$ denote robot i 's goal reference. We adopt the quadratic CLF

$$V_i(x_i) \triangleq \|x_i - x_i^{\text{goal}}(t)\|^2, \quad (17)$$

and a nominal goal-seeking command $u_i^{\text{nom}}(t)$. Robot i then computes a minimally modifying control via a quadratic program:

$$u_i^*(t) = \arg \min_{u_i, \sigma_i} \|u_i - u_i^{\text{nom}}\|^2 + \rho \sigma_i^2 \quad (18)$$

$$\text{s.t. } \dot{V}_i(x_i) \leq -cV_i(x_i) + \sigma_i \quad (19)$$

$$(15) \quad \forall j \in \mathcal{S}_i(t), \quad (16) \quad \forall j \in \mathcal{T}_i(k), \quad (20)$$

$$\|u_i\| \leq u_{\max}. \quad (21)$$

The relaxation σ_i preserves feasibility when hard constraints dominate; ρ penalizes relaxation. Crucially, connectivity CBFs are enforced only for the *active* neighbor set $\mathcal{T}_i(k)$, so the number of connectivity constraints scales with the pruned topology rather than the dense disk graph.

IV. CONNECTIVITY PRESERVATION GUARANTEE

We establish that the maintained coordination subgraph stays connected for all time. The guarantee is hybrid: (i) continuous-time link feasibility is enforced by staleness-robust connectivity CBF constraints on the maintained edges, and (ii) discrete-time pruning preserves graph connectivity by constructing a spanning tree from one-hop neighbor messages at each update.

A. Continuous-time maintenance of maintained edges via CBF

Lemma 1. Fix any undirected edge set $\mathcal{E}_c \subseteq \mathcal{V} \times \mathcal{V}$ and define

$$\mathcal{C}_{\text{conn}}(\mathcal{E}_c) \triangleq \left\{ x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \leq R_{\max}, \forall (i, j) \in \mathcal{E}_c \right\}.$$

Under the dynamics (2) and Assumptions 1–4, suppose that for all $t \geq 0$ and every $(i, j) \in \mathcal{E}_c$, the applied controls satisfy the staleness-robust connectivity inequality

$$2z_{ij}^\top(u_i - \hat{u}_j) \leq \alpha_c(h_{ij}^{\text{conn}}(x)) - 2L_f r_{ij}^2 - 4\bar{d}r_{ij} - 2r_{ij}\bar{e}, \quad (22)$$

where $z_{ij} = x_i - x_j$, $r_{ij} = \|z_{ij}\|$, $h_{ij}^{\text{conn}}(x) = R_{\max}^2 - r_{ij}^2$, and $\hat{u}_j$ and $\bar{e}$ are defined in Assumption 3. Then $\mathcal{C}_{\text{conn}}(\mathcal{E}_c)$ is forward invariant: if $x(0) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_c)$, then $x(t) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_c)$ for all $t \geq 0$.

Proof: Let $e_{ij}(t) \triangleq u_j(t) - \hat{u}_j(t)$, so $\|e_{ij}(t)\| \leq \bar{e}$ by Assumption 3. Then

$$2z_{ij}^\top(u_i - u_j) = 2z_{ij}^\top(u_i - \hat{u}_j) - 2z_{ij}^\top e_{ij} \leq 2z_{ij}^\top(u_i - \hat{u}_j) + 2r_{ij}\bar{e}.$$

TABLE I

SIMULATION SCENARIOS AND STRESS FACTORS USED IN EVALUATION.

Scenario	N	$R_{\max}$ (m)	Δt (s)	Purpose
A	10	3.0	0.05	Baseline
B	10	3.5	0.10	Temporal discretization
C	10	2.0	0.03	Sparse-connectivity stress
D	20	3.0	0.05	Scalability
E	12	2.2	0.08	Extreme stress test

Combining with (22) yields

$$2z_{ij}^T(u_i - u_j) \leq \alpha_c(h_{ij}^{\text{conn}}(x)) - 2L_f r_{ij}^2 - 4\bar{d}r_{ij},$$

which is the robust connectivity CBF condition for the true applied neighbor control. Standard CBF arguments then imply forward invariance of $\mathcal{C}_{\text{conn}}(\mathcal{E}_c)$.

B. Connectivity preservation under pruning updates

Let t_k denote the k -th topology update time. At time t_k , the graph layer outputs the maintained edge set $\mathcal{E}_{\text{topo}}(k)$ via the parent-selection and keep rule (10)–(12).

Theorem 2 (Connectivity preservation). *Assume the initial disk graph $\mathcal{G}_{t_0} = (\mathcal{V}, \mathcal{E}_{t_0})$ is connected and Assumption 4 holds so that the hop-distance routine (8) converges at each update. At each update time t_k , robots prune locally to keep exactly the incident edges in $\mathcal{E}_{\text{topo}}(k)$ defined by (12). Between updates, each robot enforces the staleness-robust connectivity CBF constraints (16) for all maintained neighbors, i.e., for all $(i, j) \in \mathcal{E}_{\text{topo}}(k)$ on $[t_k, t_{k+1})$.*

Then for every k : (i) the maintained graph $(\mathcal{V}, \mathcal{E}_{\text{topo}}(k))$ is connected at time t_k ; and (ii) all maintained edges satisfy $\|x_i(t) - x_j(t)\| \leq R_{\max}$ for all $t \in [t_k, t_{k+1})$. Consequently, end-to-end connectivity is preserved for all $t \geq t_0$.

Proof: At time t_k , Theorem 1 implies that $(\mathcal{V}, \mathcal{E}_{\text{topo}}(k))$ is a spanning tree of $\mathcal{G}_{t_k}$, hence it is connected.

Fix k . By Lemma 1 applied with $\mathcal{E}_c = \mathcal{E}_{\text{topo}}(k)$, enforcing (16) implies $\|x_i(t) - x_j(t)\| \leq R_{\max}$ for all $(i, j) \in \mathcal{E}_{\text{topo}}(k)$ and all $t \in [t_k, t_{k+1})$. Therefore each maintained edge remains present in the disk graph throughout $[t_k, t_{k+1})$. Since the maintained graph is connected at t_k and no maintained edge is lost before t_{k+1} , end-to-end connectivity persists on $[t_k, t_{k+1})$. Repeating for all k establishes the claim for all $t \geq t_0$.

V. SIMULATIONS AND RESULTS

A. Simulation setup

We evaluate the proposed distributed spanning-tree pruning and CLF–CBF controller in a $10 \times 10 \text{ m}^2$ 2D workspace with a spatially varying flow field. Robots evolve under the discrete-time approximation of (2),

$$x_i[k+1] = x_i[k] + \Delta t \left(f_{\text{flow}}(x_i[k], k\Delta t) + u_i[k] + d_i[k] \right), \quad (23)$$

with bounded input $\|u_i\| \leq u_{\max}$ and bounded disturbance $\|d_i\| \leq \bar{d}$. The communication graph is the disk graph (6)–(7) with range $R_{\max}$. We run five scenarios and compare four maintained-topology strategies:

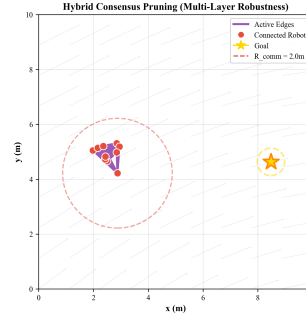


Fig. 2. Illustrative snapshot showing maintained topology: the fleet advances toward the goal while the pruning layer maintains a sparse, connectivity-feasible subgraph.

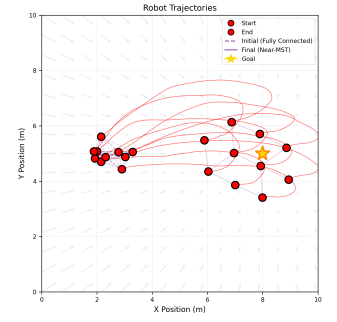


Fig. 3. Robot trajectories (Scenario A) in the flow field with topology evolving from initially dense (dashed) to sparse near-tree (solid).

- **Fully Connected:** no pruning, $\mathcal{E}_{\text{topo}}(k) = \mathcal{E}_{t_k}$.
- **Centralized MST:** an offline centralized MST (used only as a comparison oracle).
- **Adjacency Consensus:** a sequential pruning baseline (edge removals applied iteratively).
- **Distributed Pruning (ours):** the distributed hop-distance routine (8) followed by spanning-tree pruning (10)–(12).

a) *Topology-update execution:* At each topology update time t_k , robots run (8) for $q = 0, \dots, n-1$ rounds using one-hop message exchange, compute $\delta_i = \omega_{i,r}$, select parent pointers via (10), and prune locally using (11)–(12). This produces a maintained spanning tree with $|\mathcal{E}_{\text{topo}}(k)| = N-1$ edges. Between topology updates, robots enforce connectivity CBF constraints only on maintained neighbors $\mathcal{T}_i(k)$ and collision-avoidance constraints on $\mathcal{S}_i(t)$.

b) *Evaluation protocol:* We run a total of 40 simulations of 5 scenarios $\times$ 4 methods $\times$ 2 Monte-Carlo trials, each for 10 s with sampling at 0.1 s (100 evaluation points).

B. Metrics

A run is considered successful if the maintained graph remains connected for the full duration (verified by a connectivity check on $\mathcal{G}_{\text{topo}}(t)$). We report:

- **Edge reduction (%):** $((E_0 - E_f)/E_0) \times 100\%$.
- **Edge-count trajectory $E(t)$** over time.
- **Offline diagnostic connectivity margin:** minimum algebraic connectivity.

Table I summarizes five test scenarios with varying parameters to validate robustness: Scenario A, B (large time step $\Delta t=0.2$ s), C (tight communication radius $R_{\text{comm}}=2.0$ m), D (high robot count $n=20$), and E (combined extreme parameters).

VI. RESULTS

a) *Qualitative behavior:* Fig. 2 and Fig. 3 shows representative trajectories in the flow field for Scenario A. The proposed method maintains only a sparse spanning-tree structure while the team disperses and moves toward the goal.

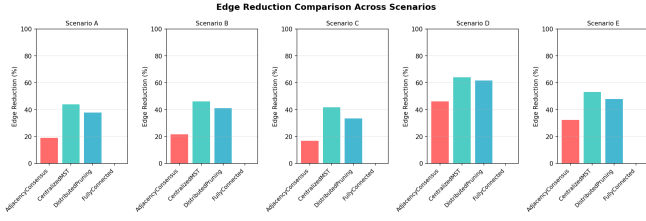


Fig. 4. Edge reduction comparison across scenarios for four topology strategies.

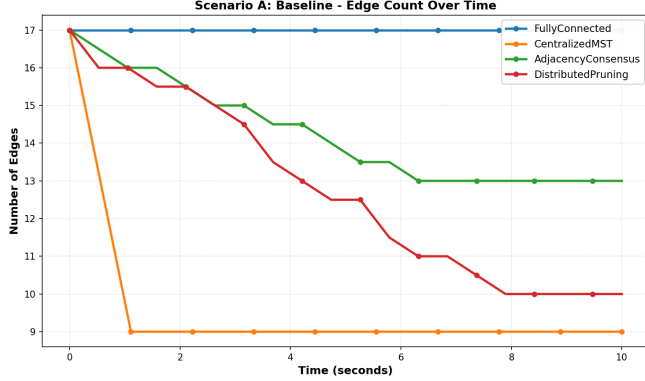


Fig. 5. Scenario A: maintained edge count $E(t)$ over time for four topology strategies.

b) *Edge reduction across scenarios.*: Fig. 4 and Fig. 6 compares final edge reduction across scenarios. Across all runs, DistributedPruning achieves strong edge reduction while preserving connectivity: on average 72.0% edge reduction (mean across trials), close to the centralized MST (80.2%) and higher than the AdjacencyConsensus baseline (67.5%).

c) *Edge-count evolution over time.*: Fig. 5 shows edge-count evolution in Scenario A, while Fig. 6 summarizes the edge-count time series for DistributedPruning across all five scenarios. DistributedPruning transitions from the initially dense topology to a near-tree structure quickly and remains sparse thereafter.

d) *Aggregate statistics (all scenarios).*: Across all 40 experiments, DistributedPruning achieves 97.5% success, with $72.04\% \pm 14.2\%$ edge reduction (mean $\pm$ std). The centralized MST is slightly less reliable (92.0%) but provides a strong sparsity comparison point. We additionally report $\min_t \lambda_2$ as an offline diagnostic connectivity margin.

TABLE II
AGGREGATE PERFORMANCE ACROSS ALL SCENARIOS

Method	Success (%)	Edge red. (%)	$\min_t \lambda_2$	Runtime (s)
Distributed Pruning (Our)	97.83	72.04 ± 14.17	0.7276 ± 1.1044	3.73 ± 4.95
Adjacency Consensus	98.21	67.20 ± 15.47	0.8297 ± 0.9205	12.91 ± 17.68
Full Graph	98.78	25.48 ± 20.18	2.0786 ± 1.1935	1.33 ± 0.77
Centralized MST	92.02	80.57 ± 4.24	0.1500 ± 0.0727	1.00 ± 0.62

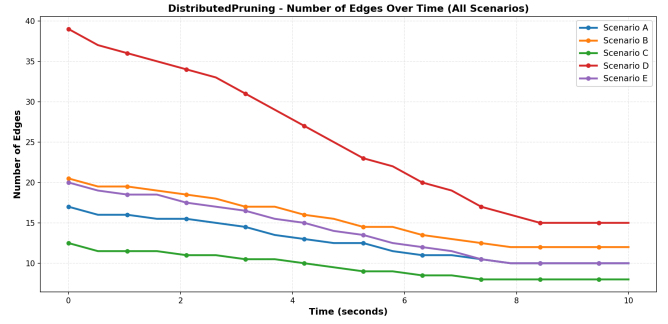


Fig. 6. DistributedPruning: maintained edge count $E(t)$ over time across scenarios.

VII. CONCLUSION

We presented a fully distributed framework for connectivity-preserving deployment of flow-driven robot fleets under limited communication. At discrete update times, robots exchange only neighbor messages to compute hop-distance estimates and prune the dense disk graph to a maintained spanning-tree coordination structure. A distributed CLF-CBF safety filter enforces collision avoidance and maintains the range constraints of the maintained links under bounded disturbances and bounded neighbor-control staleness. The resulting hybrid system preserves end-to-end connectivity while substantially reducing the number of maintained edges and associated communication constraints.

Simulations across multiple network scenarios demonstrate that the proposed distributed pruning achieves strong edge reduction while maintaining high success rates, and performs competitively relative to centralized and sequential method baselines. Future work will (i) incorporate explicit packet-drop models and fully asynchronous update protocols, (ii) extend evaluation to 3D flows, and (iii) validate the framework on field robotic platforms operating in real currents.

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